

Cross attenⁿ
same as
self attenⁿ
= 16384

FFNN
 $d_m \rightarrow d_f \rightarrow d_m$
 $64 \times 256 + 256 \times 64 = 32768$

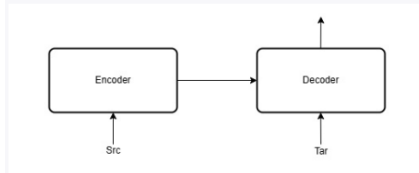
LLM Week 2

GA 2

Layer Norm
 $2 \times d_m = 128$
 $E \rightarrow 2, d \rightarrow 3$
256 384

Consider the transformer-based encoder-decoder model used for machine translation

with the following configuration (except the source and target vocabulary size, all other configuration values are the same for the encoder and decoder). The size of the vocabulary includes all the special tokens as well.



Source vocabulary size, $|V_s|$: 1000
Target vocabulary size, $|V_t|$: 1500
The context length, T : 128
The embedding dimension, $d_{embed} = d_{model}$: 64
The projection dimensions, $d_q = d_k = d_v$: 16
The number of heads, n_h : 4
The hidden dimension of FFNN, d_{ff} : 256
Uses layer norm and residual connection

We pass the sample from the source language "I enjoyed the movie Transformer" to the encoder and its translation "Naan transformer padaththai rasithen" to the decoder. All the learnable parameters are randomly initialized. The model uses the teacher-forcing approach throughout the training process.

1) Assume the model has two layers ($N = 2$). Calculate the total number of parameters in the model (excluding the embedding layer and output layer). Moreover, no bias was added to the neuron in the FFNN layers.

230656

2) How many parameters does the output layer have?

96000

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 96000

1 point

3) How many parameters do the input embedding layer have?

64000

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 64000

1 point

4) At time step $t = 1$, the prediction probability for the words "Nann", "transformer" and "padaththai" are 0.55, 0.15 and 0.2, respectively. What is the probability for the word "rasithen"? If you think, the given information is insufficient enter -1

-1

Yes, the answer is correct.

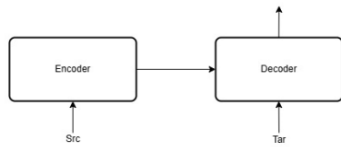
Score: 1

Accepted Answers:

(Type: Numeric) -1

1 point

$0.55 + 0.15 + 0.2 = 0.9$
The remaining 0.1 is
probability distribuⁿ over
 $1500 - 3 = 1497$ words.
But we don't know much of
0.1 is assigned rasithen.



5) The question is the continuation of the last question from the previous week].

The input embeddings for the words "learning", "brings" and "joy" are $h_1 = [0.5, 0.25, 1]$, $h_2 = [0.1, 0.25, 0]$, and $h_3 = [0.1, 0.1, 0.9]$, respectively. Note that the embeddings are row vectors. The projection matrices are as follows $W_Q = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}$ $W_K = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ $W_V = \begin{bmatrix} 0 & 0 \\ -1 & -1 \\ 1 & 1 \end{bmatrix}$

The following quantities are computed as $Q = HW_Q$ $K = HW_K$ $V = HW_V$. Let e_j denote the unnormalized attention score, a_j denote the normalized attention score (ignore the scaling by $\sqrt{d_k}$) and z_j denote the linear combination of the value vectors for the j -th word. Suppose the gradient vector $\frac{\partial L}{\partial a_1} = [1, 1]$, then what is the gradient vector $\frac{\partial L}{\partial e_1}$? Enter the sum of gradients.

Note: Keep only two significant digits after the decimal point in all your calculation

0

6) Which of the following is (are) the hyper-parameters?

1 point

- ☒ vocabulary size
- ☒ (d_k, d_v, d_q)
- ☒ number of heads
- ☒ warm-up step
- ☒ d_{model}
- ☒ number of layers

☐ size of the attention mask

7) The table below shows the input samples and the corresponding translation produced by the model.

Source	Target
I enjoyed the movie Transformers	நான் டிரான்ஸ்பார்மர்ஸ் இரைப்படத்தை ரசித்தேன்
We live among its people now	நாம் இப்பொழுது அவர்களிடையே வாழ்கிறோம்
They have much to learn	அவர்கள் கற்றுக்கொள்ள நிறைய இருக்கிறது
Freedom is the right of all sentient beings	சுதந்திரம் என்பது அனைத்து உணர்வுள்ள உயிரினங்களின் உரிமை

To produce the complete translation for the third sample in the table, we would have to run the decoder of the model at least ____ times.

Note: Ignore the [End] token.

4